Joint SAGIN Resource Allocation and Multi-Agent Path Finding for Digital-Twin-Supported Remote Robotic Applications

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Abstract—Remote multi-robot missions require both computation support and timely coordination information in environments where terrestrial infrastructure may be unavailable. This paper develops a space–air–ground integrated network (SAGIN) framework for digital-twin-supported multi-agent path finding. Robot digital twins execute PANAMA in the cloud and deliver robot-specific guidance, priorities, goal states, and intended path segments through a Drone-BS and a low-Earth-orbit backhaul. PANAMA priorities are also converted into bounded resource weights for computation allocation without modifying the priorities generated by the application. For fixed offloading decisions and radio rates, priority-aware processor allocation is obtained from KKT conditions, with a closed-form latency-only special case and explicit latency floors. To capture the application-level effect of delayed information, timestamped digital-twin states are represented by receiver caches and source-time age of information. A directed priority-inconsistency metric is then defined only for predicted vertex or edge conflicts inside PANAMA's square field of view. An age condition is established for recovery of the current PANAMA ordering, and the resulting coordination delay is bounded. The framework exposes a closed cross-layer relationship from PANAMA priority to resource allocation, information freshness, and multi-robot coordination.

Index Terms—space–air–ground integrated networks, digital twins, multi-agent path finding, age of information, task offloading, priority-aware resource allocation.

I. Introduction

Teams of mobile robots are increasingly considered for search and rescue, inspection, defence, and mining missions. These environments may lack reliable terrestrial communication and computing infrastructure. A space–air–ground integrated network (SAGIN) can provide temporary access through a low-altitude Drone-BS and extend cloud connectivity through a low-Earth-orbit (LEO) satellite. The same architecture introduces transmission, propagation, and computation delays that affect both task completion and the freshness of the information used for robot coordination.

Digital twins provide a suitable execution environment for computationally demanding multi-agent coordination. In PANAMA, asymmetric observations are created by exposing each robot to the communicated intentions of higher-priority robots within its field of view (FoV) [1]. In the original formulation, these application quantities are immediately available to the observation constructor. In a remote deployment, however, PANAMA may be executed by cloud-hosted robot digital twins and its outputs must be delivered to the physical robots. A robot may then compare stale versions of its own priority and a peer priority, which can reverse the intended yielding relation. Conventional SAGIN offloading models do not capture this consequence because they optimize communication and computation metrics without representing the coordination decision that consumes the delivered information.

This paper addresses that gap through the following contributions:

- A SAGIN architecture is formulated in which cloud-hosted robot digital twins execute PANAMA and deliver timestamped guidance and coordination states to physical robots.
- PANAMA priorities are mapped to bounded resource weights. For fixed offloading decisions and radio rates, Drone-BS and cloud CPU allocations are obtained, with a closed-form latency-only special case and a dynamic-energy KKT solution.
- A receiver-dependent priority-inconsistency metric is introduced. The metric is restricted to predicted vertex and edge conflicts inside PANAMA's square FoV and is explicitly coupled to source-time age of information (AoI).
- A communication-to-coordination relationship is established through an exact AoI condition for a current priority ordering and conservative bounds on stale decision epochs and additional waiting time.

The remainder of the paper is organized as follows. Section II presents the SAGIN, task, and information models. Section III formulates priority-aware resource allocation and derives its main latency properties. Section IV integrates delayed digital-twin information with PANAMA and develops the inconsistency analysis. Section V gives the event-driven learning model and evaluation methodology. Section VI concludes the paper.

II. System Model

A. SAGIN and Task Model

Let $\mathcal{N} = \{1, \ldots, N\}$ denote the robots. Robot i generates a task

$$\mathcal{T}_i = \{s_i, c_i\}, \tag{1}$$

where s_i is the logical input size and c_i is the number of CPU cycles per input bit. The total required work is $C_i = s_i c_i$. The mission study imposes no hard task deadline;

τ_i is retained only in the separate analytical feasibility discussion and solver validation tests. A binary variable x_i selects Drone-BS processing when $x_i = 1$ and ground-cloud processing through the LEO backhaul when $x_i = 0$. The assigned computation rates are $f_{D,i}$ and $f_{C,i}$.

B. Air-to-Ground Access

The air-to-ground model follows the probabilistic line-of-sight model in [2]. Let h_D be the Drone-BS altitude and let r_i be the horizontal distance between robot i and the Drone-BS. The probability of a line-of-sight connection is

$$P_{\mathrm{LoS}}(h_D, r_i) = \frac{1}{1 + a\exp\left[-b\left(\frac{180}{\pi}\tan^{-1}\left(\frac{h_D}{r_i}\right) - a\right)\right]}, \tag{2}$$

where a and b depend on the deployment environment. The average path loss in dB is

$$L_i^{\mathrm{G2A}} = 20\log_{10}\left(\sqrt{h_D^2 + r_i^2}\right) + z_1 P_{\mathrm{LoS}}(h_D, r_i) + z_2, \tag{3}$$

where

$$z_1 = \eta_{\mathrm{LoS}} - \eta_{\mathrm{NLoS}}, \qquad z_2 = 20\log_{10}\left(\frac{4\pi f_c}{c_0}\right) + \eta_{\mathrm{NLoS}}. \tag{4}$$

Here, f_c is the carrier frequency, c_0 is the speed of light, and η_{LoS} and η_{NLoS} are the excessive losses for the two propagation states. For bandwidth B_i^{G2A} and transmit power P_i, the uplink rate is

$$R_i^{\mathrm{G2A}} = B_i^{\mathrm{G2A}}\log_2\left(1 + \frac{P_i 10^{-L_i^{\mathrm{G2A}}/10}}{N_0 B_i^{\mathrm{G2A}}}\right), \tag{5}$$

where N_0 is the noise power spectral density.

C. Air-to-Space Backhaul

Let $\bar{d}_{\mathrm{sat}}$ be the average Drone-BS-to-satellite distance during an association interval. The air-to-space path loss is

$$L^{\mathrm{A2S}} = 20\log_{10}\left(\frac{4\pi f_{\mathrm{sat}}\bar{d}_{\mathrm{sat}}}{c_0}\right) + \xi_{\mathrm{atm}}, \tag{6}$$

where f_{sat} is the carrier frequency and ξ_{atm} is atmospheric attenuation. With bandwidth B_i^{A2S}, Drone-BS transmit power P_D, and channel gain $\|g_i\|^2$, the rate is

$$R_i^{\mathrm{A2S}} = B_i^{\mathrm{A2S}}\log_2\left(1 + \frac{P_D\|g_i\|^2 10^{-L^{\mathrm{A2S}}/10}}{N_0 B_i^{\mathrm{A2S}}}\right). \tag{7}$$

D. Task Latency

The latency for Drone-BS execution is

$$T_{i,D} = \frac{s_i}{R_i^{\mathrm{G2A}}} + \frac{C_i}{f_{D,i}}. \tag{8}$$

where $f_{D,i}$ is the Drone-AP computation capacity allocated to $\mathcal{T}_i$ in terms of CPU cycles per second. For cloud execution,

$$T_{i,C} = \frac{s_i}{R_i^{\mathrm{G2A}}} + \frac{s_i}{R_i^{\mathrm{A2S}}} + T_{\mathrm{prop}} + \frac{C_i}{f_{C,i}}, \tag{9}$$

where $T_{\mathrm{prop}} = 2[(\bar{d}_{\mathrm{sat}} + \bar{d}_{\mathrm{gw}})/c_0 + d_{\mathrm{ground}}]$ includes the outbound and return space legs and the terrestrial contribution. The approximation $2\bar{d}_{\mathrm{sat}}/c_0$ applies only if gateway and terrestrial propagation are omitted. The task latency is $T_i = x_i T_{i,D} + (1 - x_i)T_{i,C}$. The analytical expression neglects small result serialization; the simulator retains actual response serialization, contention, guard intervals, interruptions and changes in processor sharing.

E. Energy Model

The transmission energy expended by robot i is

$$E_i^{\mathrm{tx}} = P_i\frac{s_i}{R_i^{\mathrm{G2A}}}. \tag{10}$$

With fixed powers and rates, this robot-upload term is constant within a scheduling snapshot. It is excluded from the optimization objective and retained in physical energy measurements. The dynamic-voltage-and-frequency-scaling model gives the Drone-BS energy

$$E_{D,i} = x_i\kappa_D f_{D,i}^2 C_i + (1 - x_i)P_D\frac{s_i}{R_i^{\mathrm{A2S}}}, \tag{11}$$

where κ_D is an assumed switched-capacitance coefficient. Static processor power, cooling, propulsion, optical pointing, cloud electricity and downlink transmitter energy are outside this model; the reported energy is explicitly the modeled robot upload, drone backhaul and drone dynamic CPU energy.

F. End-to-End Information Freshness

PANAMA is executed by the robot digital twins in the cloud. The information delivered to a physical robot may therefore already be stale when the corresponding digital-twin message is generated. This staleness originates from the robot-to-digital-twin uplink, waiting and processing at the digital-twin layer, and the subsequent digital-twin-to-robot downlink as shown in Fig. 2. Let

$$U_j(k) = (q_j(k), g_j(k), \mathcal{L}_j(k)) \tag{12}$$

denote the kth physical-state report of robot j, where $q_j(k)$ is its sensed position, $g_j(k)$ is its goal status, and $\mathcal{L}_j(k)$ contains any additional information required by the digital twin, such as locally sensed information, experience tuples, robot's operational information, and updated models. The report is sampled at time $t_{j,k}^{\mathrm{sense}}$ and reaches the digital twin at

$$t_{j,k}^{\mathrm{DT-arr}} = t_{j,k}^{\mathrm{sense}} + \delta_{j,k}^{\mathrm{UL}}, \tag{13}$$

where $\delta_{j,k}^{\mathrm{UL}}$ is the corresponding uplink delay. The digital twin uses the received physical-state reports to generate the coordination state

$$S_j(n) = (I_j(n), \rho_j(n), g_j(n)), \tag{14}$$

where $I_j(n)$ is the intended path segment and $\rho_j(n)$ is the priority generated by PANAMA. Each downlink message m contains a payload ψ_m, a generation time t_m^{gen}, a source-information time t_m^{src}, and a sequence number ν_m. The

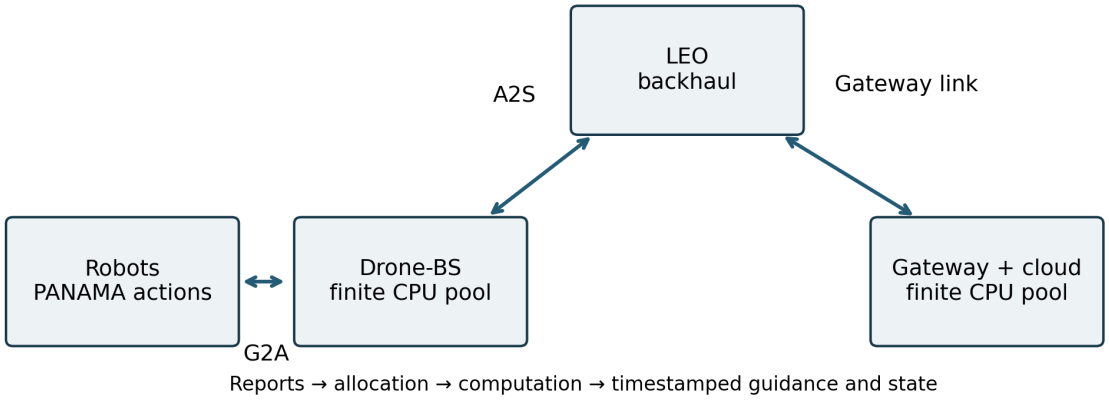


Fig. 1. SAGIN support for cloud-hosted robot digital twins and remote multi-robot coordination.

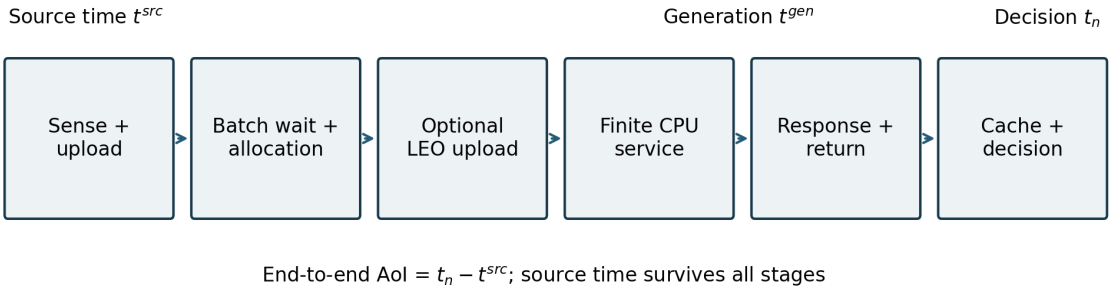


Fig. 2. System model and delay sources.

source time is not the time at which the digital twin creates the message. Instead, it records the physical sensing time of the dynamic information used to generate the payload. Let $\mathcal{D}_m$ denote the set of dynamic physical reports used to generate message m. The source-information time is defined as

$$t_m^{\mathrm{src}} = \min_{\ell \in \mathcal{D}_m} t_\ell^{\mathrm{sense}}. \tag{15}$$

Thus, $t_m^{\mathrm{src}} \le t_m^{\mathrm{gen}}$, and the difference

$$A_m^{\mathrm{pre}} = t_m^{\mathrm{gen}} - t_m^{\mathrm{src}} \tag{16}$$

captures the age accumulated before message generation, including uplink transmission, digital-twin waiting, and digital-twin processing. The message arrives at its receiver at

$$t_m^{\mathrm{arr}} = t_m^{\mathrm{gen}} + \delta_m^{\mathrm{DL}}, \tag{17}$$

where δ_m^{DL} is the delivery delay from the digital-twin layer to the physical robot. Let $\mathcal{M}_i^\sigma(t)$ denote the messages of stream σ received by robot i by time t. At decision epoch n, robot i accepts

$$m_i^{\sigma,*}(n) = \arg \max_{m \in \mathcal{M}_i^\sigma(t_n)} (t_m^{\mathrm{gen}}, \nu_m) \tag{18}$$

under lexicographic ordering. The end-to-end AoI of the accepted stream is

$$A_i^\sigma(n) = t_n - t^{\mathrm{src}}_{m_i^{\sigma,*}(n)}. \tag{19}$$

It can be decomposed as

$$A_i^\sigma(n) = \underbrace{t^{\mathrm{gen}}_{m_i^{\sigma,*}(n)} - t^{\mathrm{src}}_{m_i^{\sigma,*}(n)}}_{\text{uplink, waiting, and DT processing}} + \underbrace{t_n - t^{\mathrm{gen}}_{m_i^{\sigma,*}(n)}}_{\text{downlink delivery and cache residence}}. \tag{20}$$

The second component may be reported separately as the post-generation age

$$A_{i,\mathrm{post}}^\sigma(n) = t_n - t^{\mathrm{gen}}_{m_i^{\sigma,*}(n)}. \tag{21}$$

The primary freshness measure in this work remains the end-to-end AoI in (19), because it includes the complete physical-robot-to-digital-twin-to-robot information path.

III. Priority-Aware Resource Allocation

PANAMA priorities are application outputs rather than optimization variables. To let the infrastructure respond to those priorities, define the bounded resource weight

$$\pi_i(n) = \pi_{\min} + (\pi_{\max} - \pi_{\min}) \frac{\rho_{\max} - \rho_i(n)}{\rho_{\max} - \rho_{\min}}, \tag{22}$$

where $0 < \pi_{\min} \le \pi_i(n) \le \pi_{\max}$. The transformation preserves PANAMA's order but does not modify its priorities. The ratio $\pi_{\max}/\pi_{\min}$ controls the maximum

resource differentiation and prevents a low-priority task from receiving zero weight.

For one scheduling interval, consider

$$\min_{\boldsymbol{x},\boldsymbol{f}_D,\boldsymbol{f}_C} \quad \sum_{i=1}^{N} (\omega_2 \pi_i T_i + \omega_3 E_{D,i}) \tag{23a}$$

$$\text{s.t.} \quad \sum_{i=1}^{N} x_i f_{D,i} \le F_D^{\max}, \tag{23b}$$

$$\sum_{i=1}^{N} (1-x_i) f_{C,i} \le F_C^{\max}, \tag{23c}$$

$$f_{D,i}, f_{C,i} \ge 0, \quad x_i \in \{0,1\}, \quad i \in \mathcal{N}. \tag{23d}$$

Drone-BS platform energy can be appended when a calibrated propulsion, processing, and backhaul model is available. The formulation in (23) is a mixed-integer nonlinear program.

A. Closed-Form Computation Allocation

The square-root expressions below solve the latency-oriented special case, with zero drone CPU-energy penalty. They must not be substituted unchanged into the full latency-plus-DVFS objective. For that objective, with fixed binary assignments, the active drone rates satisfy

$$-\frac{\omega_2 \pi_i C_i}{f_{D,i}^2} + 2\omega_3 \kappa_D C_i f_{D,i} + \lambda_D = 0, \quad \lambda_D \ge 0, \tag{24}$$

with complementary slackness on the total CPU capacity. A scalar multiplier solve gives the unique positive rates for positive workloads and latency weights; capacity can remain idle. Cloud computation retains the latency-only square-root rule because cloud electricity is not included in the objective.

For fixed offloading decisions and radio rates, define $\mathcal{D} = \{i : x_i = 1\}$ and $\mathcal{C} = \{i : x_i = 0\}$. The latency-oriented Drone-BS subproblem is

$$\min_{\{f_{D,i}>0\}} \sum_{i\in\mathcal{D}} \pi_i \frac{C_i}{f_{D,i}} \quad \text{s.t.} \quad \sum_{i\in\mathcal{D}} f_{D,i} \le F_D^{\max}. \tag{25}$$

Its KKT conditions give

$$f_{D,i}^* = F_D^{\max} \frac{\sqrt{\pi_i C_i}}{\sum_{j\in\mathcal{D}} \sqrt{\pi_j C_j}}, \tag{26}$$

with the analogous cloud allocation

$$f_{C,i}^* = F_C^{\max} \frac{\sqrt{\pi_i C_i}}{\sum_{j\in\mathcal{C}} \sqrt{\pi_j C_j}}. \tag{27}$$

The resulting Drone-BS latency is

$$T_{i,D}^* = d_{i,D} + \frac{C_i}{F_D^{\max}} + \frac{\sqrt{C_i}}{F_D^{\max}\sqrt{\pi_i}} \sum_{\substack{j\in\mathcal{D}\\ j\ne i}} \sqrt{\pi_j C_j}, \tag{28}$$

where $d_{i,D} = s_i / R_i^{\mathrm{G2A}}$. Holding competing weights fixed, $\partial T_{i,D}^* / \partial \pi_i \le 0$. The minimum attainable latency under fixed rate and offloading is

$$\lim_{\pi_i \to \infty} T_{i,D}^* = d_{i,D} + \frac{C_i}{F_D^{\max}}. \tag{29}$$

The bounded mapping in (22) excludes the starvation limit $\pi_i \to 0$. For a fixed Drone-BS task set, all deadlines are feasible if and only if

$$\tau_i > d_{i,D}, \quad i \in \mathcal{D}, \qquad \sum_{i\in\mathcal{D}} \frac{C_i}{\tau_i - d_{i,D}} \le F_D^{\max}. \tag{30}$$

The same result holds for the cloud after replacing $d_{i,D}$ and $F_D^{\max}$ with $d_{i,C} = s_i/R_i^{\mathrm{G2A}} + s_i/R_i^{\mathrm{A2S}} + T_{\mathrm{prop}}$ and $F_C^{\max}$. Equations (26)–(30) characterize computation sharing.

B. Validated Binary Execution and Scheduling Relaxation

Each fixed-input snapshot is solved by enumerating the binary decisions for its at most eight new tasks. Previous unfinished tasks retain their assignments and remaining work in the capacity constraints. Each assignment uses (24) and the cloud square-root rule; the least-cost assignment is executed. Processor shares are updated on arrivals and completions, so the snapshot prediction is recorded separately from observed latency. The numerical implementation scales perspective epigraphs by their workload/objective coefficients, removes the unused cones for unavailable routes, and recovers feasible CPU shares by minimizing the same fixed-x convex subproblem. Clarabel targets 10^{-8} tolerances, with a recorded cold numerical retry at 10^{-7} if needed; accepted capacity/conic residuals and recovered-objective agreement must still satisfy 10^{-6}. A numerical retry never changes the objective or execution policy.

An independently reported continuous scheduling bound uses $x_i \in [0,1]$ and

$$a_i = x_i f_{D,i} / F_D^{\max}, \qquad b_i = (1-x_i) f_{C,i} / F_C^{\max}. \tag{31}$$

Its latency and drone energy are

$$T_i = d_{i,D} + (1-x_i)(d_{i,C} - d_{i,D}) + \frac{C_i}{F_D^{\max}} \frac{x_i^2}{a_i} + \frac{C_i}{F_C^{\max}} \frac{(1-x_i)^2}{b_i}, \tag{32}$$

$$E_{D,i} = \kappa_D C_i (F_D^{\max})^2 \frac{a_i^2}{x_i} + (1-x_i) P_D s_i / R_i^{\mathrm{A2S}}, \tag{33}$$

subject to $\sum_i a_i \le 1$, $\sum_i b_i \le 1$, $0 \le a_i \le x_i \le 1$, and $0 \le b_i \le 1 - x_i$. The closed quadratic-over-linear perspectives are represented by second-order cones, including the zero-work endpoints. CVXPY/Clarabel certifies convexity. Fractional x_i is a scheduling relaxation, not simultaneous execution of physically split jobs. We check the relaxed objective against the exact binary optimum and preserve failed snapshots without substituting another policy. Optimality applies to the scheduling snapshot, not the evolving mission.

IV. Delayed PANAMA Coordination

A. End-to-End Priority Inconsistency

Communication delay can affect the PANAMA ordering at two stages. First, delayed physical-state reports may cause the digital twin to calculate priorities from stale

robot states. Second, delayed delivery of the resulting coordination states may cause a physical robot to apply an outdated digital-twin ordering.

Let

$$\rho_j^{\mathrm{phys}}(n) \tag{34}$$

denote the priority that PANAMA would generate using the current physical state of robot j, and let

$$\rho_j^{\mathrm{DT}}(n) \tag{35}$$

denote the priority actually generated by the digital twin using the latest physical-state reports available to it.

The current physical ordering is

$$\chi_{j\prec i}^{\mathrm{phys}}(n) = \mathbf{1}\left\{\rho_j^{\mathrm{phys}}(n) < \rho_i^{\mathrm{phys}}(n)\right\}. \tag{36}$$

The digital-twin ordering is

$$\chi_{j\prec i}^{\mathrm{DT}}(n) = \mathbf{1}\left\{\rho_j^{\mathrm{DT}}(n) < \rho_i^{\mathrm{DT}}(n)\right\}. \tag{37}$$

The ordering applied by robot i is

$$\widehat{\chi}_{j\prec i}^{(i)}(n) = \mathbf{1}\left\{\widehat{\rho}_{j\to i}(n) < \widehat{\rho}_{i\to i}(n)\right\}. \tag{38}$$

The uplink-induced ordering error is

$$\Xi_{j\to i}^{\mathrm{UL}}(n) = \left|\chi_{j\prec i}^{\mathrm{DT}}(n) - \chi_{j\prec i}^{\mathrm{phys}}(n)\right|. \tag{39}$$

The post-generation delivery error is

$$\Xi_{j\to i}^{\mathrm{DL}}(n) = \left|\widehat{\chi}_{j\prec i}^{(i)}(n) - \chi_{j\prec i}^{\mathrm{DT}}(n)\right|. \tag{40}$$

The end-to-end ordering error is

$$\Xi_{j\to i}^{\mathrm{E2E}}(n) = \left|\widehat{\chi}_{j\prec i}^{(i)}(n) - \chi_{j\prec i}^{\mathrm{phys}}(n)\right|. \tag{41}$$

These metrics satisfy

$$\Xi_{j\to i}^{\mathrm{E2E}}(n) \le \Xi_{j\to i}^{\mathrm{UL}}(n) + \Xi_{j\to i}^{\mathrm{DL}}(n). \tag{42}$$

Using the FoV relevance indicator $\zeta_{j\to i}(n)$, the coordination-relevant end-to-end inconsistency is

$$\widetilde{\Xi}_{j\to i}^{\mathrm{E2E}}(n) = \zeta_{j\to i}(n)\Xi_{j\to i}^{\mathrm{E2E}}(n). \tag{43}$$

The corresponding AoI-weighted metric is

$$\widetilde{\Xi}_{j\to i}^{A,\mathrm{E2E}}(n) = \widetilde{\Xi}_{j\to i}^{\mathrm{E2E}}(n)\frac{\max\left\{A_i^{S_j}(n), A_i^{S_i}(n)\right\}}{A_{\mathrm{ref}}}. \tag{44}$$

The AoI values in (44) are measured from the robot's information generation times supporting the accepted digital-twin coordination states. They therefore include uplink delay, digital-twin waiting and processing, downlink delay, and cache residence.

V. Event-Driven Learning and Evaluation

After action n, robot i forms

$$\zeta_i(n) = \big(\widehat{o}_i(n), a_i(n), r_i(n+1), \widehat{o}_i(n+1), \mathrm{done}_i(n+1)\big), \tag{45}$$

where $\mathrm{done}_i(n+1) \in \{0,1\}$ is the terminal flag. The transition arrives at the cloud learner at $t_i^E(n) = t_{n+1} + \delta_i^E(n)$. For a sampled transition $\zeta = (o, a, r, o', \mathrm{done})$, the Double DQN target is

$$y_\zeta = r + \gamma(1-\mathrm{done})Q_{\theta^-}\left(o', \arg\max_{a'} Q_\theta(o', a')\right). \tag{46}$$

After learner update ℓ, snapshot Θ_ℓ is released at t_ℓ^{rel} and reaches robot i at $t_i^W(\ell) = t_\ell^{\mathrm{rel}} + \delta_i^W(\ell)$. Robot i installs the newest complete snapshot available before t_n. Timestamp and version checks prevent rollback.

The evaluation separates mission-time and learning-time delay. Mission experiments use a fixed policy and independently delay guidance and coordination-state delivery. Learning experiments compare immediate delivery, delayed experience uploads, delayed model delivery, and both delays. The baselines are equal resource allocation, static priority allocation, AoI-only scheduling, and PANAMA-priority-aware allocation. Recommended metrics are task latency, deadline-violation probability, energy, mission completion rate, makespan, sum of path costs, waiting steps, Ξ^{FoV}, AoI by stream, replay age, and model-version lag. Priority-weight sensitivity is evaluated through $\pi_{\max}/\pi_{\min}$, while communication sensitivity is evaluated through T_S, $\delta_S^{\max}$, blackout duration, and team size.

VI. Simulations and Results

A. Reproducible Settings and Timing

The experiments use the fixed MovingAI random-32-32-10 map, with 32 by 32 cells and one metre per cell. Robot actions are the four cardinal moves or waiting; each control cycle is one second. A square FoV of radius seven cells and five future intention cells are retained. The UAV starts at the map centre and its analytical coverage-optimal reference altitude. Its three-dimensional motion is limited to one metre per control cycle toward the maximum-coverage target computed from successfully received position reports, with a 20 m minimum altitude. No fresh simulator position is supplied to the allocation or placement controllers.

The clean frozen PANAMA checkpoint qualified on 200 episodes at each population: success rates were 97%, 96%, 95.5%, and 95% for 2, 4, 6, and 8 robots. Evaluation uses 50 separate held-out cases per population and a 250-step cap. All treatments share policy weights, starts, goals, seeds and orbital offsets. Hashes identify the checkpoint, map, case bank, channel archive and source code. Map compatibility permits exact LF/CRLF conversion only.

The 100 ms control observation budget determines which cached information is available before an action. It is

TABLE I
Nominal access, computation and timing assumptions.

Parameter	Value
G2A carrier; per-link bandwidth	2.5 GHz; 20 MHz
Urban a, b; excess LoS/NLoS losses	9.61, 0.16; 1/20 dB
Thermal PSD; receiver noise figure	-174 dBm/Hz; 7 dB
Effective receiver PSD	-167 dBm/Hz
Robot conducted uplink power	0.2 W
Access uplink/downlink service ceilings	50/50 Mbit/s
Task logical input; cycles per bit	10^5 bits; 10^3
Drone/cloud CPU capacity	$10^9/10^{10}$ cycles/s
Drone dynamic CPU coefficient	$\kappa_D = 10^{-28}$
Resource weights; completed-agent weight	$[1, 4]$; 1
Latency/energy coefficients	1 s^{-1}/1 J^{-1}
Control cycle; observation budget	1 s; 0.1 s
Maximum fragment; wire overhead	65,536 B; 48 B/fragment
Hard mission-task deadline	None

TABLE II
Backhaul engineering assumptions and full-route delays.

Parameter	Value
Uplink/downlink carrier	14.25/11.7 GHz
Uplink/downlink bandwidth	20/100 MHz
Drone conducted satellite-uplink power	2 W
Terminal EIRP; implied uplink antenna gain	35 dBW; 31.99 dBi
Satellite receive G/T	-5 dB/K
Satellite EIRP; terminal receive G/T	30 dBW; 5 dB/K
Implementation loss; Shannon efficiency	3 dB; 0.5
Atmospheric antenna diameter; exceedance	0.3 m; 1%
Nominal backhaul uplink/downlink ceilings	10/50 Mbit/s
One-way ground-network propagation	10 ms
Association-average full-route RTT	24.92–29.60 ms

not a hard task deadline. One logical coordination task is generated per robot per cycle after the initial fresh briefing. The task size and CPU demand below are simulation assumptions, not profiled neural-inference workloads or measured processor performance. Whole tasks retain their execution site once assigned; the finite processors account for existing work and recompute priority-aware sharing on arrivals and completions. CPU service is work-conserving in the latency-only case and may intentionally idle under the DVFS objective. Actual dynamic CPU energy integrates $\kappa_D f_i(t)^3$ over service.

B. Channels, LEO Geometry and Propagation

G2A uses the probabilistic urban LoS mean path loss in (3); no instantaneous small-scale fading is drawn. The thermal PSD and receiver noise figure are applied once, so the effective PSD equals -167 dBm/Hz. The rate in (5) is additionally capped by the configured shared access-service capacity. Access is time-shared with round-robin queue service; the bandwidth is not multiplied by the number of simultaneous robots. Upload input bits and actual serialized wire bits are recorded separately. To retain checkpoint compatibility, the policy's third observation channel remains PANAMA's normalized coverage-margin proxy. That proxy is distinct from the physical RF SNR used to calculate the actual access rate and is not reported as a calibrated SNR measurement.

The orbital archive contains six hours of one-second SGP4/Skyfield geometry replayed at its archived epoch, 31 August 2026. The reference site is $(46.2°, -74.6°)$ at 440 m elevation; the assumed gateway is $(45.3°, -74.6°)$ at 300 m. These coordinates are model inputs, not a measured provider gateway. The highest visible satellite is selected at 15 s scheduling boundaries, with a 25° terminal and 10° gateway elevation mask. Original 50 ms attachment/reconfiguration interruptions are retained. Frames are 20 ms with a 5% service guard.

To implement the association-average assumption, terminal and gateway ranges and statistical attenuation in dB are averaged within each association/scheduling interval. Free-space loss, SNR and Shannon rates are then recomputed from those mean inputs; arithmetic means of rates are not substituted. ITU-Rpy 0.4.0 provides the archived P.618 statistical atmospheric predictions [3] at a fixed 1% exceedance quantile. This quantile is not a sampled weather process. No provider modulation or beam-scheduler reconstruction is claimed. Small within-map UAV displacement is neglected in the orbital geometry; the conservative round-trip propagation error bound is below 0.7 microseconds, while G2A uses the actual UAV placement.

The full route is robot–UAV–LEO–gateway–ground cloud and its reverse, with no inter-satellite hop. Each direction is limited by both its terminal and gateway link budgets. Conducted transmitter power, rather than EIRP, determines transmission energy. Propagation includes one outbound and one return contribution; fragmentation does not multiply propagation by the number of fragments. Actual response serialization is simulated even though it is omitted from the analytical latency expression. Solver wall time is measured separately from logical task CPU time.

C. Validation, Baselines and Metrics

Six predictable sanity cases cover large satellite delay, fast cloud/slow drone, unavailable backhaul, unavailable drone CPU, fixed-route priority differentiation and a DVFS optimum with idle CPU. Every expected assignment is checked using an independent primal CVXPY formulation and exhaustive enumeration. The earlier supplied four-robot MATLAB validation inputs are also preserved, including the original infinite deadlines and the separate Case E feasibility test; execution here uses the approved CVXPY equivalent.

The primary grid combines uplink and common CPU multipliers $\{0.25, 1, 4\}$ under equal resource weights, episode-initial static priorities, AoI-only weights and current PANAMA-priority weights, with one fresh-coordination reference. Separate sensitivity conditions vary drone CPU, cloud CPU, downlink, logical input and cycles/bit by 0.25 and 4, the weight ratio by 1 and 16

(baseline 4), and the energy coefficient by 0, 0.1 and 10 (latency coefficient 1). There are 97 total conditions and 19,400 paired episodes. The ideal fresh reference has instantaneous coordination and no modeled support-task energy; its energy is not a hardware benchmark.

Task outcomes include observed and snapshot-predicted latency, prediction error, the 95th latency percentile, unfinished-task counts, stage service times, and modeled energy by transmitter and processor. Mission outcomes include completion, capped mission duration, successful makespan, sum of time-to-first-arrival costs, stationary steps (including blocked moves), and collision attempts. Unfinished missions remain censored at 250 steps; PAR-2 assigns them 500. Directed inconsistency is recorded with UL/DL/E2E decomposition, square-FoV relevance, source-time AoI and raw numerator/denominator counts. Mean per-episode inconsistency and pooled relevant-pair ratios are labelled separately. A zero relevance denominator gives zero.

Each comparison uses the same held-out case in both conditions. We report paired mean differences and seeded 2,000-resample 95% bootstrap intervals, with the number of usable pairs for metrics affected by censoring. The intervals are pointwise across the sensitivity grid and are not family-wise multiple-comparison guarantees. Conditional successful makespans are secondary to completion and PAR-2. Snapshot optimality and a lower relaxed objective do not imply an improvement in every mission outcome.

D. Measured Comparisons

The complete paired study is in progress. Numerical result tables and figures are generated from the finished episode records; no placeholder values are substituted.

VII. Conclusions and Future Work

A cross-layer model was developed for SAGIN-supported remote robots whose digital twins execute PANAMA in the cloud. PANAMA priorities remain application outputs, while bounded weights derived from those priorities control infrastructure resource allocation. The computation analysis identifies the priority-dependent service allocation, latency floor, and deadline-feasibility condition. The information model then connects time-stamped digital-twin states to robot caches, source-time AoI, and FoV-relevant priority inconsistencies. The resulting communication-to-coordination relationship clarifies when delayed self and peer priorities can reverse the intended ordering and provides a conservative bound on additional waiting. Future work will implement a priority- and AoI-aware radio scheduler and evaluate the framework under measured SAGIN delay and blackout traces. Furthermore, we would like to investigate the dissemination of digital twin information by relaxing the assumption that the digital twins are duplicated at the drone-AP and the cloud. Finally, we would like to improve robustness of PANAMA by incorporating stochastic priorities.

Appendix

For receiver i, let $\mathcal{V}_i(n)$ be its square radius-seven FoV and let $\widehat{I}_{i\to i}[k]$ and $\widehat{I}_{j\to i}[k]$ be the accepted self/peer intentions, padded by their last cell to horizon $H = 5$. Define vertex relevance by existence of $k \in \{0, \dots, H\}$ with equal intended cells inside $\mathcal{V}_i(n)$. Edge relevance requires a nonstationary swap at $k, k+1$ and both endpoints inside the square. The indicator $\zeta_{j\to i}$ is one when either event occurs. These are directed receiver-cache quantities; current physical priorities are used only to evaluate ordering errors. For $Q \in \{\mathrm{UL}, \mathrm{DL}, \mathrm{E2E}\}$,

$$\Xi_Q^{\mathrm{FoV}}(n) = \frac{\sum_i \sum_{j\neq i} \zeta_{j\to i}(n)\Xi^Q_{j\to i}(n)}{\max\{1, \sum_i \sum_{j\neq i} \zeta_{j\to i}(n)\}}. \tag{47}$$

The AoI-weighted form replaces each numerator term with its product by $\max\{A_i^{S_j}, A_i^{S_i}\}/A_{\mathrm{ref}}$, with $A_{\mathrm{ref}} = 1$ s. The separate numerator and denominator are stored. UL and DL errors can cancel; only the stated triangle inequality holds, so they must not be added and labelled as the E2E error.

Let $\Delta = \rho_i^{\mathrm{phys}} - \rho_j^{\mathrm{phys}}$ and $e_k = \widehat{\rho}_{k\to i} - \rho_k^{\mathrm{phys}}$ for the self and peer operands. The perceived gap is exactly $\widehat{\Delta} = \Delta + e_i - e_j$. The strict PANAMA relation is recovered if and only if $(\Delta > 0, \widehat{\Delta} > 0)$ or $(\Delta \leq 0, \widehat{\Delta} \leq 0)$. Thus, $|\Delta| > |e_i| + |e_j|$ is a sufficient, conservative margin condition.

For a fixed map and fixed goals, each reported priority is calculated from its reported state. If priority k changes by at most L_k per control cycle, source-time age gives

$$|e_k| \leq L_k \left\lceil A_i^{S_k}/T_c \right\rceil, \tag{48}$$

so an age-based sufficient condition follows by substitution in the margin inequality. The bound includes the completed-agent priority jump; a bound of one distance unit per move alone is insufficient at completion. A conservative bound is the number of grid cells per cycle. This discrete-epoch argument does not assume a continuous-time Lipschitz priority at a completion jump.

If the relevant priorities remain constant after an isolated change, reports are sampled every T_c, and sensing-to-accepted-cache delay is bounded by $D_{\max}$, both operands reflecting that change arrive within $T_c + D_{\max}$. There are then at most $\lceil (T_c + D_{\max})/T_c \rceil$ potentially stale decision epochs. This conditional bound concerns waiting due solely to that isolated stale ordering; it is not a bound on detours, repeated changes or total mission makespan under arbitrary overload.

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